

Predicting Professional CS2 Match Outcomes

A Comparative Study of Logistic Regression, Random Forest, and Gradient Boosting

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Seisenbek Dias

Problem · Data · Variables

Mergen Temirzhan

Models · Setup · Results

Onashov Aidos

Top-tier teams look identical on paper.

At the highest levels of Counter-Strike 2, aggregate statistics lose their discriminatory power. Predicting a winner before the match starts is a problem of *stochastic variation*, not just historical data.

Seisenbek Dias

02. THE EVIDENCE

7,033 648 18

Professional
Matches

Global
Tournaments

Months of Data
(May 2024 – Oct 2025)

Post-Transition Architecture: Most prior studies rely on CS:GO data. The 2023 engine transition and the HLTV 2.0 rating system effectively reset the field. This study processes only native CS2 data from the current competitive landscape to isolate genuine predictive signals.

Seisenbek Dias

03. THE VARIABLES

X1

Mean HLTV 2.0 Rating

The team's average aggregate performance score

X2

Mean Kills Per Round

The aggregate historical combat lethality

X3

Head-to-Head Win Rate

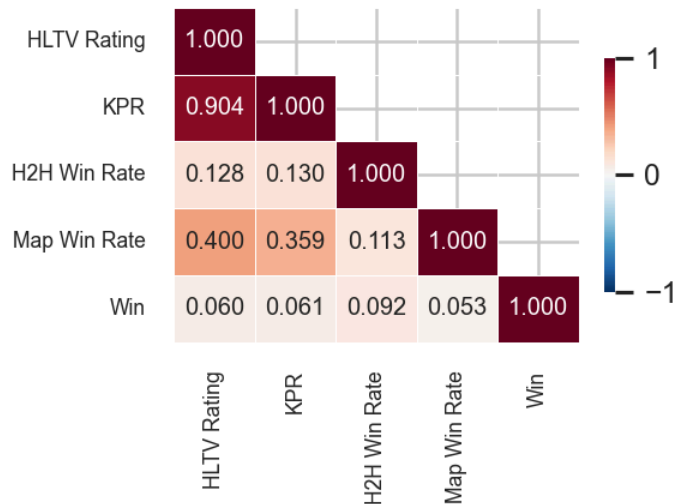
Specific historical matchup dominance between the two teams

X4

Map-Specific Win Rate

The team's historical success rate on the decided map

Pearson Correlation Matrix



* Pearson correlations: all feature-target $|r| < 0.15$ — weak individual signal

04. THE MODELS

Logistic Regression

Linear Baseline

$$P(Y=1|X) = 1 / (1 + e^{-(b_0 + b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4)})$$

Maximum likelihood with L2 regularisation. Standardised log-odds per feature SD.

Random Forest

Bagging Ensemble

$$P(Y=1|X) = (1/B) * \sum_b P_b(Y=1|X)$$

200 decision trees, majority vote. Gini-based feature importances. Robust to overfitting.

Gradient Boosting

Additive Ensemble

$$F_m(x) = F_{(m-1)}(x) + \eta * h_m(x) \mid \eta = 0.05, M=200$$

Sequential shallow trees fitting pseudo-residuals. Highest hold-out accuracy.

05. THE VERDICT

Logistic Regression

56.1%

Accuracy

F1: 0.484

AUC: 0.574

Random Forest

57.2%

Accuracy

F1: 0.536

AUC: 0.571

Gradient Boosting

56.4%

Accuracy

F1: 0.528

AUC: 0.573

Paired t-tests (5-fold CV): differences between classifiers are not statistically significant ($p > 0.05$)

The Bound of Stochastic Variation

A naïve baseline — simply predicting the majority class (always predict 'Win') — yields an automatic 54.6% accuracy.

Applying a 200-tree Random Forest model lifts that accuracy by just **2.6%**.

The models are *not failing*; they have reached the limits of pre-match data.

Mergen Temirzhan

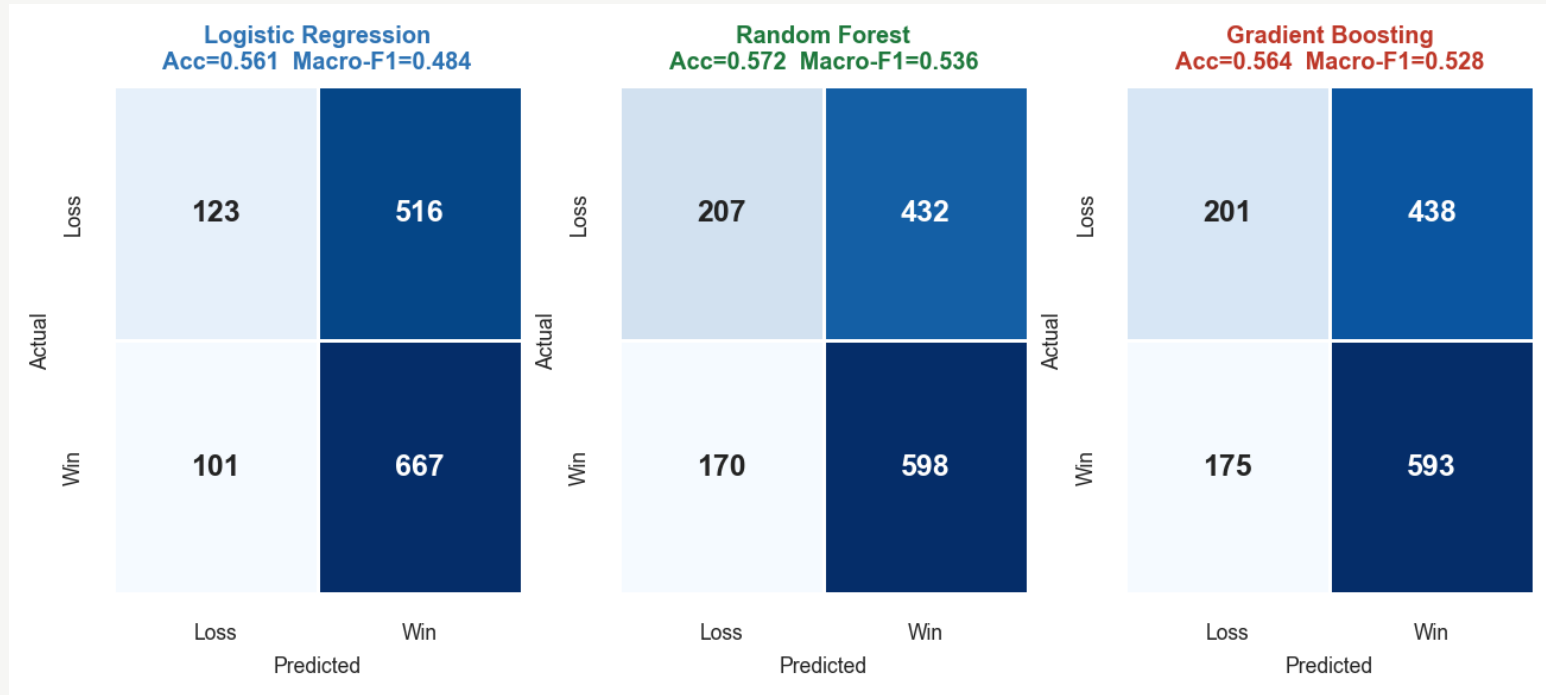
57.2%

Random Forest Maximum Accuracy

54.6%

Naïve Baseline

07. CONFUSION MATRICES



All models over-predict Win — false positive rate is the dominant error, as expected from weak feature-target correlations

01

Map Win Rate

Importance: 0.403 (RF)

The dominant predictor. At the elite tier, being pulled into an unfamiliar structural environment creates a measurable, objective disadvantage that raw skill cannot overcome.

Onashov Aidos

02

Head-to-Head

Importance: 0.269 (RF)

The secondary signal. Stylistic matchups and historical dominance persist over time, creating psychological and tactical friction between opponents.

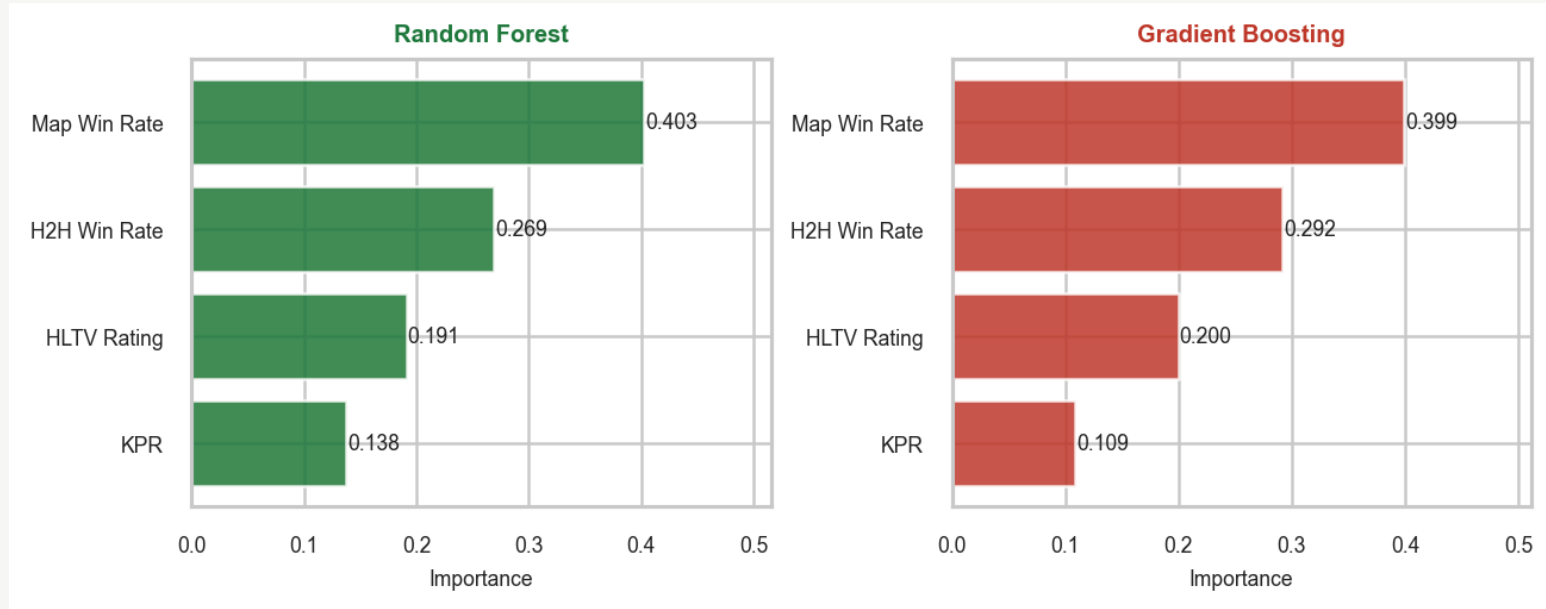
03

Aggregate Skill

Importance: 0.191 (HLTV) · 0.138 (KPR)

The weakest signal. When both teams are elite, historical combat lethality and overall ratings cancel each other out.

09. FEATURE IMPORTANCE — BOTH ENSEMBLE MODELS



Map Win Rate ranks first in both RF and GB with ~40% importance. KPR is consistently the weakest contributor.

10. THE FEATURE CEILING

Pre-Match Horizon

Historical Aggregates

Temporal Noise

Map Veto

Roster Shifts

Max Accuracy: ~56%

In-Game Reality

Real-Time Economy

Alive-Player State

Round Score

Clutch Variance

Max Accuracy: 80%+

The algorithms do not lack capacity.

The data lacks certainty.

At the highest tier of competition, pre-match outcomes are bound by a structural ceiling of stochastic variance.

To break the 56% barrier, we must look past aggregate history and measure the game in motion.

- H3 ✓ Map Win Rate is the dominant predictor
 - H2 ~ H2H Win Rate is the secondary signal
 - H1 v HLTV Rating / KPR — weakest contributors
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